# Customer Behaviour Analysis

Instacart Market Basket Analysis

## Total Orders

3.35M

## Unique Customers

206K

## Peak orders window

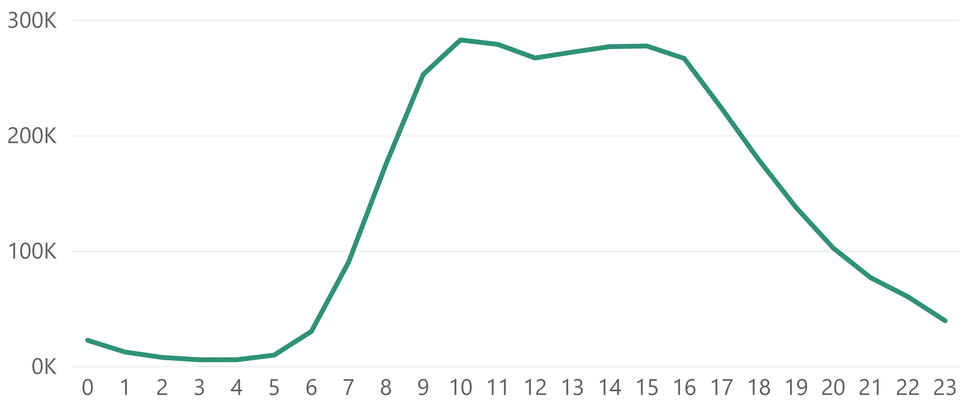
10am - 5pm

2.14M

## Weekly return orders

315.33K

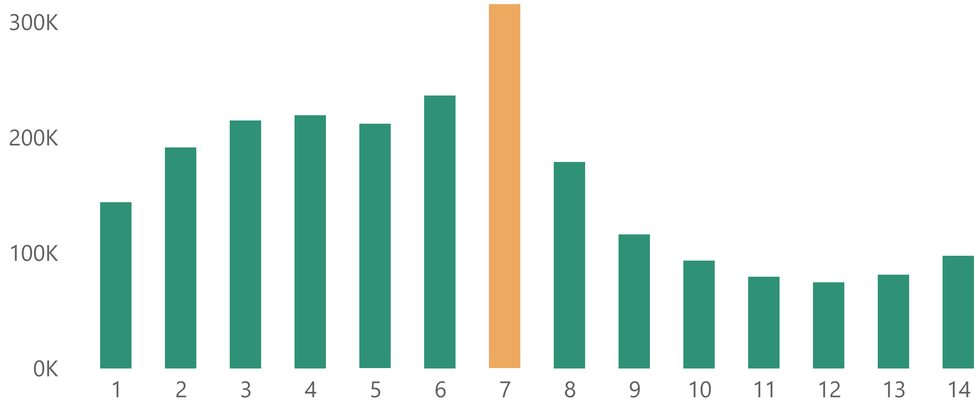
### When Do Customers Place Orders?




### Shopping Activity by Day and Hour

| order_day_of_week | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 3841 | 2336 | 1376 | 933 | 782 | 1142 | 3254 | 12097 | 27445 | 39764 | 47244 | 49711 | 50144 | 52414 | 53077 | 52538 | 48173 | |
| 1 | 3594 | 1791 | 1091 | 728 | 791 | 1581 | 5282 | 16294 | 33561 | 51027 | 54719 | 50640 | 46153 | 45832 | 45853 | 45504 | 43868 | |
| 2 | 3007 | 1543 | 928 | 709 | 731 | 1377 | 4670 | 13012 | 24227 | 35635 | 38486 | 37437 | 35018 | 35957 | 36387 | 36711 | 36762 | |
| 3 | 2870 | 1460 | 932 | 635 | 691 | 1322 | 4467 | 12157 | 22124 | 31694 | 35316 | 34543 | 32789 | 33439 | 34060 | 35208 | 34538 | |
| 4 | 2571 | 1477 | 870 | 669 | 718 | 1306 | 4313 | 12274 | 21392 | 30779 | 34305 | 33132 | 31519 | 32021 | 32930 | 33435 | 33363 | |
| 5 | 3124 | 1626 | 996 | 825 | 896 | 1530 | 4757 | 13132 | 23476 | 33503 | 37587 | 37087 | 34944 | 35514 | 36561 | 36660 | 35047 | |
| 6 | 3217 | 1870 | 1182 | 844 | 784 | 1116 | 3170 | 11066 | 22439 | 30127 | 34813 | 36066 | 36261 | 36708 | 37791 | 37151 | 34693 | |

### When do customers return for shopping?




## KEY INSIGHTS & RECOMMENDED ACTIONS

### 01 | Peak Ordering Hours

- **Insight:** Orders concentrate between 10 AM and 5 PM.
- **Action:** Compare hourly order demand with fulfillment capacity. Adjust peak-hour staffing where processing delays are observed.

### 02 | Weekly Replenishment Opportunity

- **Insight:** Seven-day gaps account for 315K orders, with a 69.39% product reorder rate.
- **Action:** Test day-6 replenishment reminders for customers with repeated weekly patterns against a control group

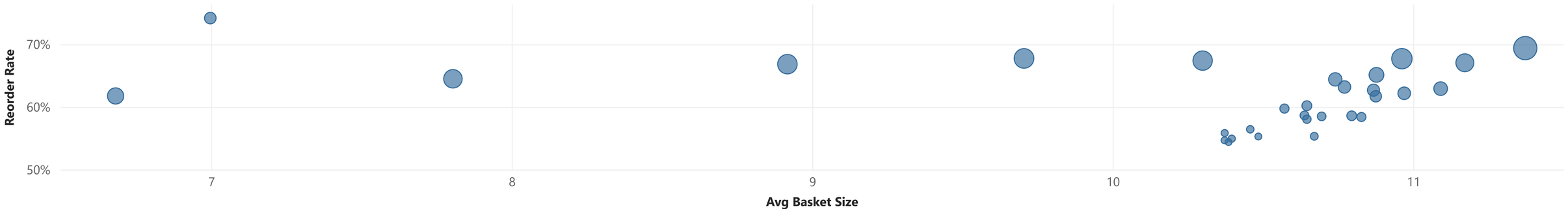
**Measure success:** Order-processing time and on-time fulfillment for peak-hour operations; incremental repeat-order rate for replenishment reminders.

# Basket and Reorders Analysis

instacart

| Product purchases | First added reorder rate | Avg Basket Size | Reorder Rate |
|---|---|---|---|
| 33.82M | 68% | 10.11 | 59.01% |

### Does Basket Size Relate to Repeat Purchasing?

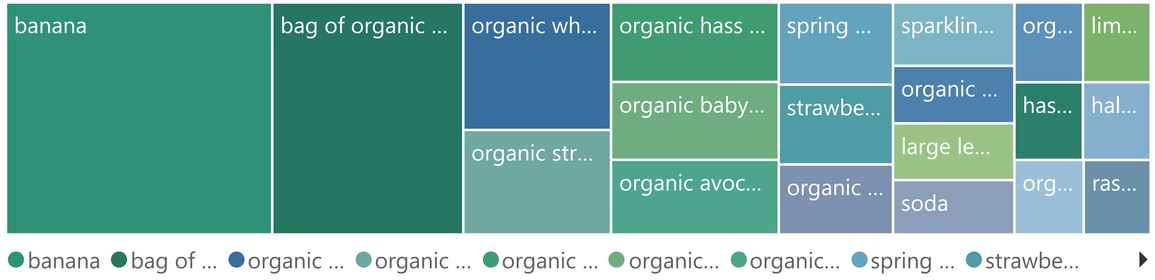



### How Does Product Familiarity Evolve Across Orders?




### Products added first in cart

Top 20 most frequently added first




## KEY TAKEAWAYS

- 68% of first-added products were previously purchased, compared with 59% across all product purchases.
  **Recommendation:** A/B test a personalized Buy Again shortcut at the start of shopping. Measure time to first cart addition and completed-order conversion.

- Orders after 7-day and 27-day gaps average 11.37 and 10.40 products respectively, but their reorder rates differ by 14.46 percentage points.
  **Recommendation:** Test recommendations tailored to purchasing history and return interval against a generic product list. Measure recommended-product add-to-cart rate.

# High Demand Products

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## Distinct Products

49.69K

## Dept with max orders

Produce

9.89M

## Product purchased highest

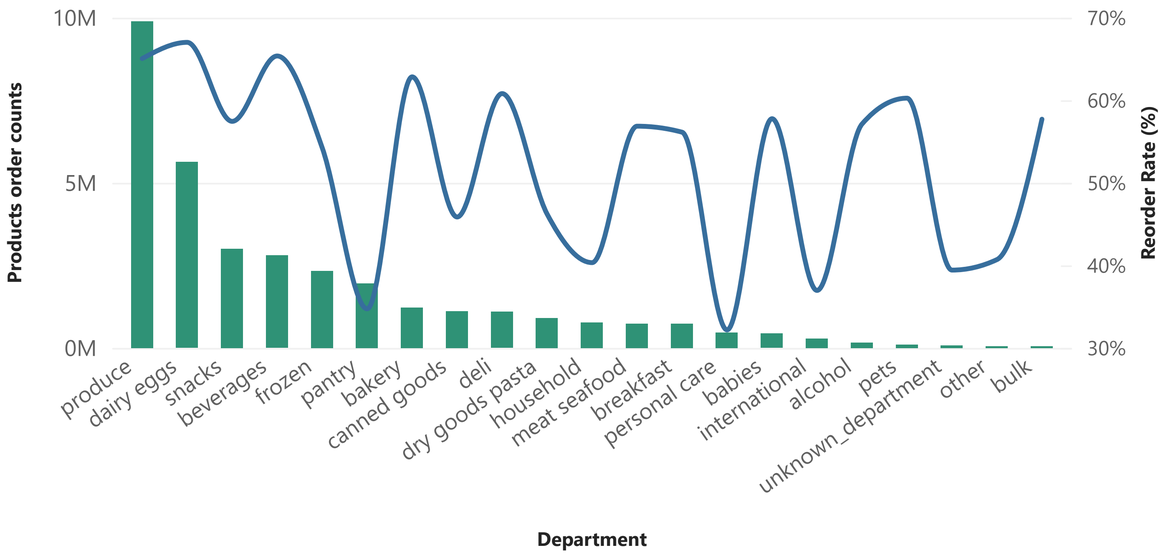
Banana

491.29K

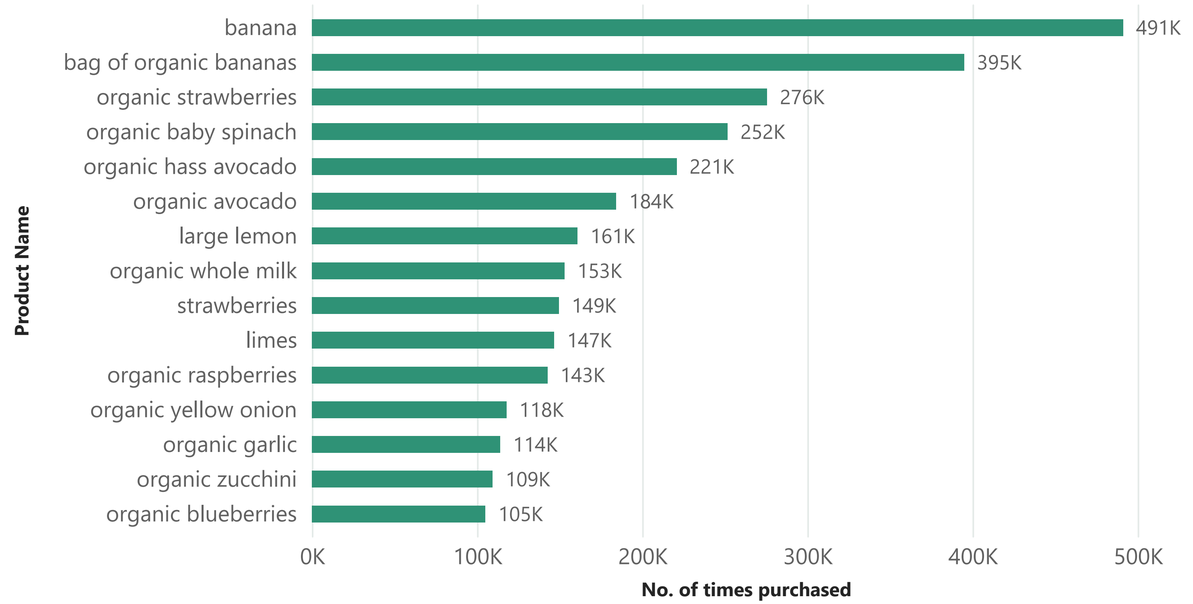
## Banana reorderd

84.51%

### Which Departments Drives highest purchases and reorders?

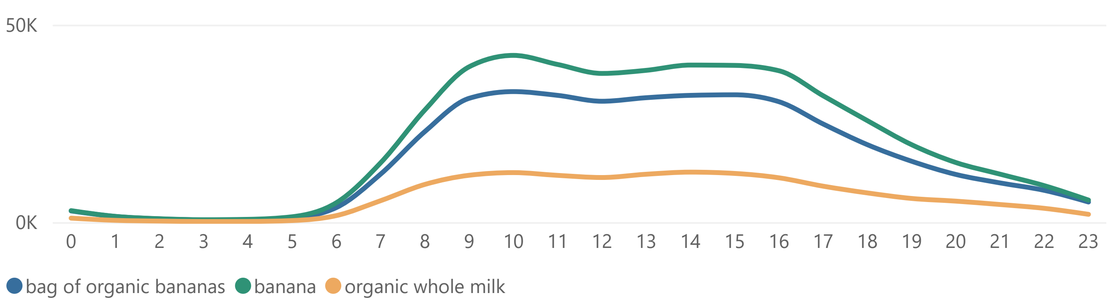



### What Are the Most Frequently Purchased Products?




### Products demand throughout day

banana ✕ organic whole milk ✕ +1




## KEY TAKEAWAYS & RECOMMENDED ACTIONS

- Produce accounts for **29.24% of all product purchases (9.89M)**, making it Instacart's most frequently purchased department.
  **Recommendation:** Prioritize the top 10 Produce products for inventory monitoring and replenishment planning, using stockout data to identify availability gaps.

- Banana was purchased **491K times**, with an **84.51% reorder rate**, making it a highly popular and frequently repurchased product.
  **Recommendation:** Test suggesting products frequently bought alongside bananas through a Frequently Bought Together feature. Measure additional products added to carts and completed purchases.